

SILICON LABS IoT CHALLENGE 2026

SMART HEALTHCARE WITH EDGE AI · SEMIFINAL SUBMISSION

TARANG

Heart Intelligence in Motion

Innovation & Differentiation Report

A Continuous Edge Intelligence Platform for Cardiac Monitoring

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Document II of the Semifinal Submission Package

Prepared for Competition Judges · Confidential

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1 Executive Summary

Tarang is a wearable cardiac-monitoring platform built on the Silicon Labs EFR32MG26, designed to close a specific gap in cardiac care: no existing category combines continuous wear, clinical-grade multi-signal acquisition, and real-time, on-device intelligence in a single low-power edge node. Consumer smartwatches offer convenience but only on-demand, single-lead snapshots. Certified Holter monitors and clinical telemetry offer diagnostic rigor but sacrifice continuity and immediacy, reporting retrospectively rather than in real time. Tarang is architected to occupy the space between them.

The platform fuses three sensing modalities – ECG, PPG, and IMU – acquired through a hardware-only, zero-CPU pipeline unique to the EFR32MG26's peripheral fabric (IADC, PRS, and LDMA), filtered in real time by an IMU-referenced adaptive noise model, and classified on-device by a quantized 1D convolutional neural network running on the SoC's Matrix Vector Processor. Results – not raw waveforms – are transmitted over an encrypted BLE 5.3 link to a gateway that renders a clinical dashboard and logs events for review.

Why It Matters

Cardiovascular disease remains the leading cause of death worldwide, and a meaningful share of that burden is driven by arrhythmias that are intermittent, asymptomatic, and consequently missed by episodic testing. Continuous, real-time, on-body classification converts detection from a matter of chance – whether the patient happens to be tested at the right moment – to a matter of course.

Why Now

Edge-AI-capable, ultra-low-power SoCs with hardware-accelerated signal acquisition and on-chip inference have only recently become available at a cost and power budget compatible with a wearable form factor. Tarang exists because the EFR32MG26 makes an architecture possible that was not practical on general-purpose microcontrollers.

Why It Is Different

Tarang's differentiation is not a single feature but a systems-integration outcome: five compounding engineering decisions – zero-CPU acquisition, multimodal fusion, adaptive artifact rejection, on-device inference, and event-driven connectivity – co-designed on one SoC without a companion processor to absorb the overflow.

Why It Is Scalable

Because acquisition costs almost no CPU budget, the same hardware has headroom for capabilities beyond arrhythmia detection – blood-pressure estimation, sleep staging, and population-scale screening – without a new SoC. The roadmap in Section 20 describes this progression.

5	1	3	0
engineered innovations	SoC, zero external MCU	sensing modalities fused	CPU cycles to acquire ECG

2 The Healthcare Challenge

Cardiovascular disease is the leading cause of death globally, responsible for an estimated 19.8 million deaths a year according to the World Health Organization – more than any other cause. A large share of these deaths are premature: among the roughly 18 million premature deaths (under age 70) caused by non-communicable disease in 2021, at least 38 percent were attributable to cardiovascular disease. The WHO characterizes most of this burden as preventable, provided disease is detected and managed early enough.

Atrial fibrillation, the arrhythmia most directly relevant to Tarang's initial clinical use case, illustrates why early detection is structurally difficult rather than merely under-resourced. Global AF prevalence rose from an estimated 33.5 million people in 2010 to roughly 59 million by 2019, and epidemiological reviews put the lifetime risk of developing AF after age 45 at approximately one in three to one in five. Because AF is frequently paroxysmal and asymptomatic, a substantial fraction of cases are never captured by a single clinical encounter: U.S. claims-based studies estimate that undiagnosed cases account for somewhere between roughly one in ten and nearly a quarter of the total AF population, and many patients are first diagnosed only after a stroke has already occurred.

This is a monitoring-architecture problem before it is a treatment problem. The tools available today were not designed to solve it.

Where Monitoring Breaks Down

- Episodic testing structurally misses intermittent arrhythmias – a 12-lead ECG or a 30-second wrist reading only captures an event if it happens to occur during the test.
- Continuity and clinical accuracy are traded against each other. Devices that monitor continuously are typically not diagnostic-grade; devices that are diagnostic-grade are typically not worn continuously.
- Remote and resource-limited settings – rural clinics, home recovery, ambulance transport – are precisely where continuous, connectivity-independent monitoring is most needed and least available.
- Retrospective review, as with a 24–48 hour Holter recording, means a clinically significant event on day one is not identified until the recording is read days later – after the patient has already gone home.

SO WHAT

The unmet need is not another wearable that measures heart rate. It is a device that stays on the body, understands what it is measuring well enough to tell signal from noise, and says something the moment it matters – not days later.

3 Why Existing Solutions Are Insufficient

The cardiac-monitoring market looks crowded from the outside: smartwatches, fitness bands, and hospital-grade Holter monitors all claim to watch the heart. Examined closely, each occupies a narrow corner of a triangle whose three vertices are continuity, clinical accuracy, and accessibility – and none occupies the center.

Consumer Smartwatches

Devices such as the Apple Watch, Samsung Galaxy Watch, and Fitbit Sense-class products popularized wrist ECG, but their capability is a single-lead, on-demand, roughly 30-second snapshot that requires the user to consciously trigger a reading and hold still. There is no continuous rhythm monitoring and no always-on arrhythmia surveillance; between snapshots, heart-rate tracking falls back to PPG alone – an optical signal that cannot reliably distinguish arrhythmia from motion noise.

Fitness Bands

Fitness bands trade clinical relevance for accessibility. They report heart rate, steps, and sleep stages from PPG and accelerometer data but carry no ECG hardware at all. Their heart-health scores are wellness metrics intended to motivate a workout, not diagnostic signals capable of catching a silent, asymptomatic episode.

Holter Monitors and Clinical Telemetry

At the other extreme, certified medical devices – Holter recorders, ICU bedside monitors, and ambulance ECG units – deliver genuine diagnostic-grade signal fidelity using the same lead configuration Tarang adopts for its proof-of-concept. But they are built for the clinic, not for daily life: wired multi-lead harnesses, non-ambulatory form factors, and offline, retrospective analysis. A Holter recording is worn for 24–48 hours and read days later by a cardiologist; if the relevant event happened on day one, the patient has already gone home unaware.

The Gap Tarang Closes

Smartwatches sacrifice clinical rigor for convenience. Clinical monitors sacrifice continuity and real-time response for rigor. Tarang is architected from the silicon up to avoid making that trade-off: continuous wear, a clinical-grade three-electrode signal, and a real-time on-device decision, in one low-power edge node.

4 Why Tarang Exists

Tarang was not conceived as a wearable with sensors added to it. It was conceived by asking what a modern edge-AI SoC makes newly possible in cardiac monitoring, and then designing the clinical use case around that answer. That ordering – platform capability first, product form second – is what separates a systems-engineering approach from a feature-accretion one, and it is why Tarang is best understood as a platform rather than a single device.

Mission

To make continuous, clinically credible cardiac monitoring available outside the hospital – on the wrist, in the ambulance, and at home – without requiring a network connection or a companion processor to make it work.

Vision

A single low-power edge node that can be trusted to watch a heart continuously, tell real signal from motion and noise, and decide – locally, in under a second – whether what it is seeing is worth a clinician's attention.

Platform Strategy, Not Product Strategy

The architectural decisions described in this report – hardware-only acquisition, multimodal fusion, adaptive filtering, on-device inference, and event-driven connectivity – were made to generalize beyond arrhythmia detection. The same pipeline that classifies a heartbeat today has the computational and structural headroom to support blood-pressure estimation, sleep-stage classification, and population-scale screening tomorrow, without new silicon. Section 20 develops this progression in detail. Tarang's near-term goal is a validated arrhythmia-monitoring device; its structural goal is a reusable cardiac-and-biosignal edge-intelligence platform.

5 Tarang Platform Overview

At a system level, Tarang is a six-stage pipeline that starts with three physical sensors and ends with a clinician-facing dashboard. The design principle running through every stage is the same: keep the computationally cheap, deterministic work in hardware, and reserve the CPU for the one thing only software can do – interpretation.

Stage	Function	Where It Runs
1. Sensing	ECG electrodes, MAX30102 PPG/SpO ₂ , MPU6050 IMU capture raw physiological and motion signals.	On-body sensors
2. Noise Cancellation	An IMU-referenced adaptive (NLMS) filter removes motion artifact from the ECG in real time.	EFR32MG26 (DSP)
3. Classification	A quantized 1D-CNN scores each beat and flags arrhythmia patterns.	EFR32MG26 (MVP accelerator)
4. Computation	Heart rate, SpO ₂ , HRV metrics, and an arrhythmia score are derived from the classified stream.	EFR32MG26
5. Transmission	Only classified events, summarized traces, and periodic vitals cross an encrypted BLE 5.3 link.	EFR32MG26 → gateway
6. Review	A timestamped clinical dashboard and the underlying clean ECG waveform are made available for review.	Raspberry Pi gateway

The property worth underlining for a non-engineering reader: stages 1 through 4 execute entirely on the wearable, with no cloud dependency and no companion processor. The BLE link in stage 5 carries a decision, not a data stream – the intelligence that determines what matters has already happened at the edge by the time anything leaves the device.

6 The Zero-CPU ECG Architecture

This is the single most defensible engineering claim in this report, and the architectural decision with the largest downstream effect on every other subsystem. On a conventional MCU-based acquisition path, the CPU configures the ADC, waits on or is interrupted by each conversion, moves samples into a buffer, and only then hands off to filtering or classification – and every one of those steps costs cycles, wakes the core, and introduces scheduling jitter.

On the EFR32MG26, Tarang instead chains three hardware peripherals directly to one another: the Incremental ADC samples the conditioned ECG signal at a fixed, hardware-timed cadence with no software polling; the Peripheral Reflex System routes each conversion-complete signal directly to the DMA engine as a hardware trigger, peripheral talking to peripheral with the CPU never in the loop; and Linked DMA moves each sample into a circular SRAM buffer automatically, chaining descriptors so a multi-second frame assembles with zero CPU intervention. The Cortex-M33 core wakes only once a full, clean frame is ready for filtering and classification.

Why This Matters, In Four Outcomes

- Power – the core stays in its lowest-power sleep state for the overwhelming majority of the duty cycle, waking only for frame-level processing and BLE events. For a continuous-monitoring device, this is the difference between hours of wear and days of wear.
- Determinism – arrhythmia detection is a timing problem: R-R interval variability and HRV metrics (RMSSD, SDNN) are only meaningful if sample timing is stable. A hardware-timed chain samples with clock-level determinism, independent of CPU load, which makes the timing-sensitive features the classifier depends on trustworthy by construction rather than by careful interrupt tuning.
- Scalability – because acquisition costs almost no CPU budget, the same core has headroom to run PPG/IMU sampling, adaptive filtering, CNN inference, and the BLE stack concurrently on one SoC. A CPU-driven acquisition design would have consumed the budget that multimodal fusion and on-device AI now use.
- Reliability – a hardware datapath cannot be pre-empted by a runaway task, a misbehaving BLE callback, or a firmware bug in an unrelated module, removing an entire class of failure modes – dropped samples, timing glitches, priority-inversion bugs – that would otherwise sit upstream of a clinically relevant decision.

SO WHAT

Zero-CPU acquisition is not an optimization layered on top of the product — it is the reason the rest of the architecture (fusion, adaptive filtering, on-device AI, and BLE, all running concurrently) fits on a single SoC's power budget at all.

7 Multimodal Sensor Fusion

ECG measures electrical activity with excellent temporal precision but says nothing about perfusion or oxygenation, and a wearable lead system is highly susceptible to motion and electrode-contact artifact with no independent signal to check itself against. PPG is convenient and requires no gel electrodes, but it measures blood-volume change at a peripheral vessel – several physiological steps removed from the heart's electrical cycle – and cannot resolve fine rhythm morphology or reliably distinguish a true arrhythmia from a noisy optical read. A PPG-only device reporting an irregular rhythm is reporting a hypothesis, not a diagnosis.

Tarang fuses three signals that fail in different, largely uncorrelated ways. ECG can be electrically clean but electrode-motion-corrupted. PPG can be optically clean but perfusion-ambiguous. IMU carries no physiological signal at all but knows, unambiguously, when the body is moving. Used together, IMU tells the system when to trust ECG and PPG less; PPG cross-validates heart rate independently of ECG electrode-contact quality; and ECG remains the ground truth for rhythm morphology. No competing wearable in the consumer tier fuses all three signals; most clinical monitors fuse none, treating each channel as an independent stream for a human to interpret later.

SO WHAT

Fusion turns three individually fallible sensors into one confidence estimate that is more robust than any single modality — and it is precisely what the adaptive filter (Section 8) and the on-device classifier (Section 9) are built to exploit.

8 The Signal Intelligence Layer

Motion artifact is the single largest source of false alarms and missed detections in ambulatory ECG. A fixed-coefficient low-pass filter, the conventional answer, removes noise indiscriminately – including real cardiac features that happen to fall in the same frequency band. Tarang instead treats the IMU as an active noise reference rather than a step-counter: an adaptive Normalized Least-Mean-Squares filter uses the accelerometer and gyroscope signal to model motion-correlated noise and subtract it from the raw ECG in real time, continuously adjusting to the specific motion the wearer is currently experiencing rather than applying a static assumption.

This sits alongside baseline-wander removal, bandpass filtering, and continuous signal-quality monitoring, which together determine, frame by frame, whether the current signal is clean enough to classify or should be flagged as low-confidence rather than risk a false read.

The Outcome, Not the Implementation

For a clinician or a judge, the relevant fact is not the filter's coefficients – it is that the classifier downstream is only ever shown signal the system has already judged trustworthy. That distinction is what separates a device that reports a heart-rate estimate from one that can defensibly flag an arrhythmia.

9 The Edge AI Layer

A quantized (INT8) 1D convolutional neural network, compiled through TensorFlow Lite for Microcontrollers and executed on the EFR32MG26's Matrix Vector Processor hardware accelerator, classifies each beat and flags arrhythmia patterns – including atrial fibrillation and PVC-type events – directly on the wearable. No raw waveform has to leave the device for a diagnosis to be produced.

Why Cloud-Only Healthcare Is Insufficient Here

- Latency – a cloud-inference architecture requires a full round trip from device to gateway to network to inference server and back before an arrhythmia can be flagged. For a condition where minutes matter, that round trip is a clinical liability, not a UX inconvenience.
- Privacy – because classification happens locally, the raw ECG waveform never has to be transmitted for a diagnosis to be produced. Only classified events, summarized traces, and periodic vitals cross the BLE link, which is both a power optimization and a meaningfully smaller privacy surface than continuous raw-biosignal streaming.
- Connectivity dependence – rural clinics, home settings with unreliable broadband, and moving ambulances are exactly the environments where a cloud-dependent monitor is least useful and most needed at the same time. Tarang's detection pipeline requires no network connection to run.
- Consistency – sub-second, offline, always-available classification means Tarang behaves identically in a hospital ward with full Wi-Fi or in the back of an ambulance with zero signal bars. That consistency, not peak-condition benchmark performance, is what real-world medical-adjacent devices are actually judged on.

10 The Clinical Intelligence Platform

Beyond the wearable itself, Tarang is an ecosystem, not isolated firmware. Rather than continuously streaming raw ECG over the air, Tarang transmits over an encrypted BLE 5.3 link only when there is something worth transmitting: a classified arrhythmia event, a summarized ECG trace, or periodic vitals. A Raspberry Pi gateway on the receiving end runs a FastAPI backend, persists events to SQLite, and renders a Next.js clinical dashboard – logging events and able to raise alerts, while the intelligence that decided what mattered already happened at the edge.

What the Dashboard Is For

The dashboard is not a data lake; it is a clinical review surface. It presents heart rate, HRV, beat classification, arrhythmia flags, and the clean ECG waveform in a timestamped, chronologically navigable form – built for a clinician deciding whether an event warrants follow-up, not for a data scientist exploring a raw signal archive.

This ecosystem view matters for evaluation purposes: judging Tarang on the wearable alone understates the project. The backend, storage, and dashboard layers exist because a monitoring device that cannot get its findings in front of a clinician in a usable form has not actually solved the clinical problem, regardless of how good its on-wrist signal processing is.

11 Why Silicon Labs

This report has been careful throughout to root every capability claim in a named peripheral rather than a generic description, and that discipline is deliberate: Tarang's architecture is not a feature set implemented on generic hardware and described in Silicon Labs terms after the fact. It is a direct, deliberate exploitation of what makes the EFR32MG26 different from a commodity Cortex-M part.

Silicon Labs Capability	What It Enables in Tarang
IADC	Fixed, hardware-timed ECG sampling with no software polling – the foundation of the zero-CPU pipeline.
PRS (Peripheral Reflex System)	Routes conversion-complete signals directly to DMA, peripheral to peripheral, with the CPU never in the acquisition loop.
LDMA (Linked DMA)	Assembles multi-second ECG frames into SRAM automatically via chained descriptors, with zero CPU intervention.
MVP Accelerator	Executes the quantized 1D-CNN beat classifier on-chip, enabling sub-second, offline arrhythmia detection.
Secure Vault (AES-256)	Underpins the encrypted BLE 5.3 link that carries only classified events and summaries, not raw waveform.
EM2/EM3 Low-Energy Modes	Let the core sleep for the overwhelming majority of the duty cycle between hardware-timed frames.
BLE 5.x Radio	Provides the event-driven, low-payload transport that generalizes from a single gateway to a fleet architecture (Section 20).

The Narrative, Not the Peripheral List

Every other section of this report – zero-CPU acquisition, concurrent multimodal fusion, on-device AI, and secure event-driven connectivity, all on one power budget – depends on this platform choice. A commodity Cortex-M SoC without this peripheral fabric would have forced a companion MCU, a larger form factor, and a smaller power budget for everything downstream of acquisition. Silicon Labs did not just make Tarang possible; it made the specific architecture in this report the correct engineering answer.

12 Competitive Landscape

The comparison below is deliberately narrow in scope: it evaluates Tarang against what a well-engineered conventional wearable does today, across the dimensions that determine whether a device is clinically trustworthy rather than merely feature-complete.

At a Glance

Before the detailed, category-by-category comparison below, the summary view: across the five compounding innovations that make up the Tarang stack (Section 2), a typical competing wearable checks few or none of the boxes Tarang checks by design.

Feature	Competitor	Tarang
Continuous ECG	Limited – on-demand snapshot only	Yes
Motion artifact rejection	No	Yes
On-device AI	No	Yes
Zero-CPU acquisition	No	Yes
Multimodal fusion	No	Yes

Category-by-Category Comparison

Dimension	Apple Watch / Galaxy Watch	Fitness Bands	Holter / Clinical Telemetry	Tarang
Monitoring continuity	On-demand snapshot	Continuous, non-diagnostic	Continuous, offline	Continuous, real-time
Edge intelligence	Limited, app-processor based	None	None (retrospective read)	On-device 1D-CNN via MVP accelerator
Motion robustness	Basic / none	Basic	N/A (worn at rest / clinic)	IMU-referenced adaptive (NLMS) filtering
Sensor fusion	None	PPG + IMU (wellness only)	None	ECG + PPG + IMU, cross-validated
Clinical workflow fit	Consumer app only	Consumer app only	Familiar to clinical staff, but retrospective	Familiar lead topology + real-time alerting

Every check mark a competing device earns in a feature comparison is typically true – and that is precisely the point. On paper, spec sheets converge. The differentiation lives in implementation depth: how continuity is achieved without draining a battery in hours, how artifact is rejected without discarding real cardiac signal, and how a classification decision is made in under a second without a network round trip. Sections 6 through 10 exist to substantiate each row above with a named mechanism, not a marketing claim.

13 Differentiation Matrix

Capability	Conventional Wearable	Tarang
ECG	Single lead, on-demand	Continuous, 3-electrode clinical topology
PPG	HR / SpO ₂ estimate only	Fused with ECG and IMU context
IMU	Optional, steps / activity	Used as an active noise reference
Sensor fusion	Limited or none	ECG + PPG + IMU, cross-validated
Motion-artifact removal	Basic fixed-coefficient filtering	NLMS-adaptive, IMU-referenced
Edge AI	Limited or cloud-dependent	On-device INT8 1D-CNN via MVP accelerator
Acquisition model	CPU-driven (interrupt / poll loop)	Zero-CPU (IADC + PRS + LDMA hardware path)

Every claim in the right-hand column is traceable to a specific peripheral, algorithm, or design decision named earlier in this report – not to a marketing adjective. That traceability is itself part of the differentiation: it is what allows a judge, a clinician, or a technical reviewer to verify the claim rather than take it on faith.

14 The Tarang Innovation Moat

Defensibility in a hardware-software medical-adjacent platform does not come from a single patent-able trick; it comes from the difficulty of replicating an entire co-designed stack under the same power and cost constraints. Tarang's moat is built in five compounding layers, each of which raises the cost of a competitor replicating the layer above it.

Layer	What Makes It Hard to Replicate
1. Hardware architecture	The zero-CPU IADC → PRS → LDMA acquisition chain is a specific, non-obvious use of EFR32-class peripherals; a competitor without deep familiarity with this peripheral fabric would default to a conventional CPU-driven design and inherit its power and jitter costs.
2. Sensor fusion	Fusing three signals that fail independently requires a validated cross-referencing model, not just three sensors on a board – the value is in how IMU is used to modulate trust in ECG and PPG, not in the parts list.
3. Signal intelligence	The adaptive NLMS filter must be tuned to reject motion noise without attenuating true QRS morphology – a calibration problem that requires real physiological data, not a generic DSP library default.
4. Edge AI	Training and quantizing a classifier to INT8 for the MVP accelerator without an unacceptable loss of sensitivity is a non-trivial ML-engineering problem, and one that improves with more validated data over time.
5. Clinical platform	The backend, storage, and dashboard layers turn a signal into a clinical workflow artifact – replicating the wearable alone does not replicate the ability to get a finding in front of a clinician usably.

SO WHAT

Any single layer in this table is a reasonable capstone project on its own. The moat is that all five had to be co-designed on one power and timing budget, on one SoC — which is difficult to shortcut, and gets harder to catch up to as more validated data and firmware iterations accumulate.

15 Commercial Opportunity

Tarang's near-term commercial path follows the same logic as its technical architecture: start where the clinical need is most acute and the deployment friction is lowest, then extend outward. This section deliberately avoids projecting inflated addressable-market figures; the opportunity is described in terms of the segments and adoption logic that follow from the platform's actual capabilities.

Target Users, In Order of Near-Term Fit

- Post-cardiac-event patients recovering at home, who need extended monitoring without hospital tethering – the segment with the clearest clinical justification for continuous, real-time monitoring today.
- Remote and rural clinics, where connectivity-independent, on-device detection is not a convenience but a precondition for the device to function at all.
- Ambulance and pre-hospital emergency care, where the RA/LA/RL gel-electrode topology Tarang adopts is already the familiar standard, lowering training burden for paramedics to effectively zero.
- Preventive and screening-oriented use, where continuous background monitoring reframes arrhythmia detection from an event that requires the patient to be tested at the right moment to one that is caught whenever it actually occurs.

Adoption Logic

Each of these segments shares a common decision driver: a clinician or care system choosing Tarang is choosing it because the alternative – episodic testing or hospital-bound telemetry – structurally fails to catch what they need to catch, not because Tarang is a cheaper or more convenient version of an existing category. That is a slower sales motion than a consumer-wearable go-to-market, and it is the correct one for a device making clinically relevant claims; hospitals, clinics, and remote-care providers adopt monitoring tools based on demonstrated reliability, not marketing velocity.

16 Clinical Impact

Earlier Detection

Continuous background monitoring converts arrhythmia detection from a matter of chance to a matter of course, directly addressing the paroxysmal, asymptomatic character of conditions like atrial fibrillation described in Section 2.

Continuity of Care Beyond the Hospital Wall

A patient recovering from a cardiac event does not need to remain tethered to a hospital bed to be monitored safely – they need a device that behaves like one. Tarang's continuous, on-device classification is designed to let clinicians extend monitoring beyond the hospital wall without extending clinical risk.

Access in Under-Resourced Settings

Because the detection pipeline runs entirely at the edge, Tarang is architected to work identically in a district clinic as in a tertiary hospital – the settings without reliable connectivity or nearby specialist access are exactly where a cloud-dependent monitor is least useful and most needed.

Workflow Efficiency

By transmitting only classified events and summaries rather than continuous raw waveform, and by presenting findings through a clinician-facing dashboard rather than a raw data feed, Tarang is designed to reduce the volume of undifferentiated data a clinical team has to review to find what matters.

A Note on Claims

Tarang's clinical-impact case in this report is architectural and mechanistic – what the design is built to enable – rather than a claim of measured clinical outcomes. Section 17 sets out precisely what has and has not yet been validated, and Section 19 states the current limitations plainly.

17 Validation Status & Deployment Pathway

Tarang is a semifinal-stage engineering platform, and this report is deliberately explicit about what that does and does not mean. The subsystems below have each been independently validated; full end-to-end, on-body validation and clinical validation are the next stages, not completed work.

Current Status

Status	Scope
Validated	ECG acquisition (zero-CPU pipeline) · PPG acquisition · IMU acquisition · BLE communication · backend architecture · dashboard architecture
In progress	Full system integration · PCB bring-up · end-to-end on-body validation · AI model optimization

Building Beyond the Prototype

Validating each subsystem independently is a necessary but not sufficient condition for a clinically usable device. The path from where Tarang stands today to deployment readiness runs through five stages, and Team Ocelleon is deliberately not claiming to have completed stages beyond the first.

Stage	Focus
1. Engineering validation	Verify each subsystem – acquisition, filtering, classification, connectivity – performs to specification in isolation. (Current stage.)
2. Signal validation	Confirm signal quality and artifact rejection hold up across real-world motion and use conditions, on-body and integrated.
3. Algorithm validation	Evaluate classifier sensitivity and specificity against established reference datasets and, subsequently, prospectively collected data.
4. Clinical pilot	Structured evaluation with clinical partners and informed participants, under appropriate ethical and regulatory oversight.
5. Deployment readiness	Hardening, certification pathway engagement, and manufacturing readiness ahead of any real-world clinical deployment.

SO WHAT

Tarang is presented here as what it is: a rigorously validated engineering platform at Stage 1, with a clear-eyed and specific pathway — not a vague promise — to the stages that would precede real-world clinical use.

18 Responsible AI, Governance & Security

Human-in-the-Loop Design

Tarang is designed to assist clinicians, not to replace them. The on-device classifier's role is to surface a candidate event for clinical review – through the dashboard described in Section 10 – not to issue an autonomous diagnosis or trigger treatment. Every arrhythmia flag is intended to be a prompt for clinical judgment, not a substitute for it.

Model Transparency and Governance

- Confidence scoring – classification outputs are designed to carry a confidence signal alongside the flag itself, rather than a bare binary alert, so low-confidence reads (e.g., from a degraded signal) can be distinguished from high-confidence ones.
- Traceability – every classification is intended to be traceable back to the underlying signal window and quality assessment that produced it, so a clinician reviewing an event can see what the model saw.
- Dataset governance – model development draws on established, well-characterized public arrhythmia datasets, including MIT-BIH, INCART, and PTB-XL, used for training and evaluation groundwork; this is a starting point for model development, not a claim of clinical-grade validation on Tarang's own hardware signal.
- Ongoing validation – as the platform progresses through the stages in Section 17, model performance is intended to be re-evaluated against increasingly representative, prospectively collected data.

Future Security Architecture

Tarang's connectivity layer already uses an encrypted BLE 5.3 link. The items below are near-term security roadmap commitments, presented as roadmap rather than current claims:

- Authenticated BLE pairing to prevent unauthorized device association.
- End-to-end encrypted communications from sensor to dashboard, not only over the radio link.
- Secured backend transport between the gateway and any downstream storage or analytics layer.
- Audit logging of data access and event review for accountability.
- Role-based access control for clinical dashboard users.

Standards Considered During Development

Team Ocelleon has developed Tarang with awareness of, and future alignment toward, the following standards and frameworks. This is a statement of awareness and design intent, not a claim of current certification or compliance – formal certification against any of these frameworks is outside the scope of a competition-stage prototype and would follow, not precede, the clinical validation stages in Section 17.

- ANSI/AAMI EC57 – the reference framework for testing arrhythmia-detection algorithm performance.
- EU AI Act – relevant to the classification and governance of AI systems used in health-adjacent contexts.
- HIPAA – relevant to protected health information handling in any eventual U.S. clinical deployment.
- GDPR – relevant to personal and health data handling in any eventual EU deployment.

19 Current Limitations

A credible engineering report states what is not yet true as clearly as what is. The following are Tarang's current, honest limitations at the semifinal stage:

- Full PCB integration of all sensing and processing subsystems onto a single board is still in progress; validated subsystems have not all yet been proven together on final hardware.
- Clinical validation – evaluation against real patient data under clinical oversight – has not yet been conducted; current model development draws on public reference datasets (Section 18).
- The security roadmap described in Section 18 is a set of near-term commitments, not a currently implemented, audited security posture.
- Pilot studies with clinical partners have not yet begun.
- Regulatory certification (e.g., under FDA or EU MDR pathways) is outside the scope of this competition project and has not been pursued at this stage.

Why This Section Is Here

Stating limitations plainly is not a hedge against the rest of this report — it is what makes the rest of it credible. A submission that claimed clinical validation, security compliance, or regulatory readiness it had not earned would undermine the legitimate, verifiable engineering claims made in Sections 6 through 11.

20 Roadmap & Scalability

The architecture described in this report was deliberately built with computational and structural headroom for capabilities beyond arrhythmia detection alone. None of the items below require a new SoC – they require new models and firmware modules running on the pipeline already built.

Direction	What It Builds On
Blood-pressure estimation	Pulse transit time, derivable from the timing offset between the ECG R-peak and the PPG waveform Tarang already fuses, is a well-established path toward cuffless BP estimation – a modeling effort on sensors already on the device.
Sleep monitoring	The IMU and PPG channels already captured for motion-artifact rejection and perfusion sensing are the same inputs sleep-stage classifiers rely on – an additional inference head on the same MVP accelerator, not new hardware.
Population-scale AFib screening	The BLE-to-gateway architecture generalizes directly from a single Raspberry Pi dashboard to a fleet of gateways feeding an aggregate screening pipeline; the event-driven, low-payload protocol was designed with that scale in mind from the start.
Cloud analytics	The gateway is a natural aggregation point for longitudinal, multi-patient trend analysis – a layer that sits above, and does not have to touch, the on-device real-time detection path.
Personalized thresholds	On-device inference plus a persistent per-patient event log is the foundation for adaptive, patient-specific thresholds rather than population-average ones.
Dry / textile electrodes	The electrode-agnostic analog front end validated with gel electrodes in this proof-of-concept is the direct predecessor to a dry-electrode or conductive-textile version for an always-on wristband or chest-strap form factor – the form factor originally envisioned for Tarang.

The proof-of-concept's 3-electrode gel configuration – the same RA/LA/RL, right-leg-drive topology used in hospital bedside monitors and ambulance ECG units – was a deliberate validation choice, not a step back from the original wearable vision. It lets the team prove the zero-CPU acquisition and on-device AI pipeline against a signal quality clinicians already trust, on hardware compatible with how paramedics and nurses already work, while dry and textile electrode integration remains an explicit near-term roadmap item rather than an open engineering question.

21 Team Ocelleon

Tarang's engineering surface spans six domains – analog hardware, firmware and peripheral orchestration, digital signal processing, machine learning, wireless protocol design, and full-stack software – and its core difficulty is that none of them could be solved in isolation: every decision in one domain constrained every other. Delivering the validated subsystems described in this report required the team to co-design across that full span concurrently, on a single power and timing budget, rather than treating each domain as an independent workstream handed off in sequence.

Domain	What It Required
Hardware	Analog front-end design and right-leg-drive circuit tuning for a clinical-grade instrumentation amplifier signal chain.
Firmware	Orchestrating IADC, PRS, and LDMA into a zero-CPU acquisition chain, with correct power-state transitions between sleep and processing.
DSP	Tuning an adaptive filter to reject motion-correlated noise without attenuating true cardiac signal.
Machine learning	Training and quantizing a 1D-CNN for an on-chip accelerator without an unacceptable loss of classification sensitivity.
Connectivity	Designing an event-driven, encrypted BLE protocol that transmits only clinically meaningful payloads.
Full-stack software	Backend, storage, and a clinician-facing dashboard that turn a classification into a usable clinical workflow.

That breadth – not any single component – is the execution capability this report asks judges to evaluate: the same team that got a hardware-timed acquisition chain working also built the model that runs on it, the protocol that reports it, and the dashboard that displays it.

22 Executive Investment Thesis

Problem

Cardiovascular disease is the leading cause of death worldwide, and a substantial share of its burden – particularly from arrhythmias like atrial fibrillation – is driven by conditions that are intermittent and frequently asymptomatic, and therefore structurally missed by episodic testing. No existing product category combines continuous wear, clinical-grade signal acquisition, and real-time on-device intelligence.

Solution

Tarang is a wearable edge-intelligence platform, built on the Silicon Labs EFR32MG26, that acquires ECG through a zero-CPU hardware pipeline, fuses it with PPG and IMU data, removes motion artifact adaptively, and classifies arrhythmia on-device – transmitting only clinically meaningful results to a clinician-facing dashboard.

Differentiation

The defensibility is architectural, not feature-level: five compounding innovations co-designed on a single SoC, without a companion processor, at a power budget that makes continuous multi-day wear viable. Section 14 details why this is difficult to replicate quickly.

Market Direction

Adoption begins in segments where continuous, connectivity-independent, real-time monitoring is not a convenience but a functional requirement – post-cardiac-event home recovery, remote and rural clinics, and pre-hospital emergency care – and extends outward as the platform's headroom is used for adjacent capabilities described in Section 20.

Team Capability

The team has independently validated acquisition, filtering, classification, connectivity, and full-stack software across six technical domains, demonstrating the cross-disciplinary execution a platform of this complexity requires.

Why Now

Edge-AI-capable SoCs with hardware-accelerated acquisition and on-chip inference at wearable-compatible power budgets have only recently become available – the architecture in this report was not practical on the previous generation of general-purpose microcontrollers.

Why Tarang

Tarang is presented honestly, at its actual stage of maturity: a validated engineering platform with a specific, credible pathway – not a vague promise – toward clinical validation and deployment. That combination of technical depth and stated humility about what remains unproven is, itself, part of the investment case.

23 Closing Statement

Tarang was not built by adding sensors to a wearable. It was built by asking what a Silicon Labs Series 2 SoC makes uniquely possible – zero-CPU acquisition, hardware-accelerated inference, secure low-power connectivity – and then designing the clinical use case around that answer.

Why Tarang. The gap between convenient wearables and rigorous clinical monitors is real, well-documented, and, until recently, difficult to close on a single low-power SoC. Tarang closes it.

Why now. The silicon that makes this architecture possible – hardware-timed acquisition, on-chip inference, secure connectivity, all on one power budget – has only recently become available.

Why Team Ocelleon. Delivering validated subsystems across hardware, firmware, DSP, machine learning, connectivity, and full-stack software required exactly the cross-disciplinary execution this report has described, and the team has demonstrated it.

Why Silicon Labs. Every claim in this report is traceable to a specific EFR32MG26 peripheral, accelerator, or capability – this is a platform-native architecture, not a generic design retrofitted with a Silicon Labs label.

Why advancement is justified. Tarang is a rigorously validated engineering platform, honest about what remains unproven, with a specific pathway toward clinical credibility – not simply a wearable prototype, but an emerging intelligent cardiac-monitoring platform with a defensible path toward real-world healthcare impact.

TARANG – Heart Intelligence in Motion

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